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To whom it may concern,

I am writing this letter of recommendation at the request of Mr. Olivier Ndinga Oba, whom I supervised while he completed his Master's thesis at IRIDIA, the artificial intelligence laboratory in the Université libre de Bruxelles.

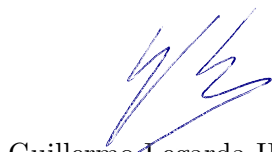
During our time working together, Olivier demonstrated outstanding qualities as a researcher. His thesis was focused on the use of topological methods for imitation learning in the context of swarm robotics. Despite initially having no knowledge of algebraic topology and only limited experience in the design of robot swarms, Olivier adapted almost immediately, quickly reaching the point where we could have discussions as peers, not just as student and supervisor. He carried out his research with remarkable diligence, discipline, and professionalism, ensuring there was always an open channel of communication between us and always keeping me updated on his progress. He was also capable of working independently, constantly bringing in original ideas that would shape the results of his work, which he would accurately interpret and build upon. In short, it was an absolute pleasure working with him.

From a technical perspective, Olivier demonstrated strong coding proficiency, easily switching between C++ and Python codebases, and translating Python to C++ when necessary. His work also required him to use specific libraries for vector-field analysis (natural Helmholtz-Hodge decomposition), persistent homology calculations (Gudhi) and heuristic optimisation (i-race). At no point did Olivier seem intimidated by the breadth of topics he needed to familiarise himself with. Instead, he was able to understand and integrate all these libraries into an efficient framework, producing valuable tools for our research group.

In conclusion, I can offer nothing but the highest praise for Mr. Olivier Ndinga Oba. His work ethic surpasses any I have encountered in a Master's student, and I am confident that he has the technical and professional expertise to become an outstanding researcher.

Should you have any further questions, please, do not hesitate to get in touch with me.

Yours faithfully,



Guillermo Legarda Herranz
PhD student